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java.time.chrono

Class MinguoChronology

java.lang.Object
    java.time.chrono.AbstractChronology
        java.time.chrono.MinguoChronology

All Implemented Interfaces:

`Serializable`, `Comparable<Chronology>`, `Chronology`

```
public final class MinguoChronology
extends AbstractChronology
implements Serializable
```

The Minguo calendar system.

This chronology defines the rules of the Minguo calendar system. This calendar system is primarily used in the Republic of China, often known as Taiwan. Dates are aligned such that `0001-01-01 (Minguo)` is `1912-01-01 (ISO)`.

The fields are defined as follows:

- era - There are two eras, the current 'Republic' (ERA_ROC) and the previous era (ERA_BEFORE_ROC).
- year-of-era - The year-of-era for the current era increases uniformly from the epoch at year one. For the previous era the year increases from one as time goes backwards. The value for the current era is equal to the ISO proleptic-year minus 1911.
- proleptic-year - The proleptic year is the same as the year-of-era for the current era. For the previous era, years have zero, then negative values. The value is equal to the ISO proleptic-year minus 1911.
- month-of-year - The Minguo month-of-year exactly matches ISO.
- day-of-month - The Minguo day-of-month exactly matches ISO.
- day-of-year - The Minguo day-of-year exactly matches ISO.
- leap-year - The Minguo leap-year pattern exactly matches ISO, such that the two calendars are never out of step.

Implementation Requirements:

This class is immutable and thread-safe.

Since:

1.8

See Also:

Serialized Form

Field Summary

Fields

Modifier and Type	Field	Description
static MinguoChronology	INSTANCE	Singleton instance for the Minguo chronology.

Method Summary

Concrete Methods

Modifier and Type	Method
MinguoDate	date(Era era, int yearOfEra, int month, int dayOfMonth)
MinguoDate	date(int prolepticYear, int month, int dayOfMonth)
MinguoDate	date(TemporalAccessor temporal)
MinguoDate	dateEpochDay(long epochDay)

MinguoDate	**dateNow**()
MinguoDate	**dateNow**(**Clock** clock)
MinguoDate	**dateNow**(**ZoneId** zone)
MinguoDate	**dateYearDay**(**Era** era, int yearOfEra, int dayOfYear)
MinguoDate	**dateYearDay**(int prolepticYear, int dayOfYear)
MinguoEra	**eraOf**(int eraValue)
List<**Era**>	**eras**()
String	**getCalendarType**()
String	**getId**()
boolean	**isLeapYear**(long prolepticYear)

ChronoLocalDateTime<MinguoDate>	localDateTime(TemporalAccessor temporal)
int	prolepticYear(Era era, int yearOfEra)
ValueRange	range(ChronoField field)
MinguoDate	resolveDate(Map<TemporalField,Long> fieldValues, ResolverStyle resolverStyle)
ChronoZonedDateTime<MinguoDate>	zonedDateTime(Instant instant, ZoneId zone)
ChronoZonedDateTime<MinguoDate>	zonedDateTime(TemporalAccessor temporal)

Methods inherited from class java.time.chrono.AbstractChronology

compareTo, equals, hashCode, toString

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Methods inherited from interface java.time.chrono.Chronology

from, getDisplayName, period

Field Detail

INSTANCE

```
public static final MinguoChronology INSTANCE
```

Singleton instance for the Minguo chronology.

Method Detail

getId

```
public String getId()
```

Gets the ID of the chronology - 'Minguo'.

The ID uniquely identifies the `Chronology`. It can be used to lookup the `Chronology` using `Chronology.of(String)`.

Specified by:

`getId` in interface `Chronology`

Returns:

the chronology ID - 'Minguo'

See Also:

`getCalendarType()`

getCalendarType

```
public String getCalendarType()
```

Gets the calendar type of the underlying calendar system - 'roc'.

The calendar type is an identifier defined by the *Unicode Locale Data Markup Language (LDML)* specification. It can be used to lookup the `Chronology` using `Chronology.of(String)`. It can also be used as part of a locale, accessible via `Locale.getUnicodeLocaleType(String)` with the key 'ca'.

Specified by:

`getCalendarType` in interface `Chronology`

Returns:

the calendar system type - 'roc'

See Also:

getId()

date

```
public MinguoDate date(Era era,
                       int yearOfEra,
                       int month,
                       int dayOfMonth)
```

Obtains a local date in Minguo calendar system from the era, year-of-era, month-of-year and day-of-month fields.

Specified by:

date in interface Chronology

Parameters:

era - the Minguo era, not null

yearOfEra - the year-of-era

month - the month-of-year

dayOfMonth - the day-of-month

Returns:

the Minguo local date, not null

Throws:

DateTimeException - if unable to create the date

ClassCastException - if the era is not a MinguoEra

date

```
public MinguoDate date(int prolepticYear,
                       int month,
                       int dayOfMonth)
```

Obtains a local date in Minguo calendar system from the proleptic-year, month-of-year and day-of-month fields.

Specified by:

date in interface Chronology

Parameters:

prolepticYear - the proleptic-year

month - the month-of-year

dayOfMonth - the day-of-month

Returns:

the Minguo local date, not null

Throws:

DateTimeException - if unable to create the date

dateYearDay

```
public MinguoDate dateYearDay(Era era,
                              int yearOfEra,
                              int dayOfYear)
```

Obtains a local date in Minguo calendar system from the era, year-of-era and day-of-year fields.

Specified by:

dateYearDay in interface Chronology

Parameters:

era - the Minguo era, not null

yearOfEra - the year-of-era

dayOfYear - the day-of-year

Returns:

the Minguo local date, not null

Throws:

DateTimeException - if unable to create the date

ClassCastException - if the era is not a MinguoEra

dateYearDay

```
public MinguoDate dateYearDay(int prolepticYear,
                              int dayOfYear)
```

Obtains a local date in Minguo calendar system from the proleptic-year and day-of-year fields.

Specified by:

dateYearDay in interface Chronology

Parameters:

prolepticYear - the proleptic-year

dayOfYear - the day-of-year

Returns:

the Minguo local date, not null

Throws:

DateTimeException - if unable to create the date

dateEpochDay

public MinguoDate dateEpochDay(long epochDay)

Obtains a local date in the Minguo calendar system from the epoch-day.

Specified by:

dateEpochDay in interface Chronology

Parameters:

epochDay - the epoch day

Returns:

the Minguo local date, not null

Throws:

DateTimeException - if unable to create the date

dateNow

public MinguoDate dateNow()

Description copied from interface: Chronology

Obtains the current local date in this chronology from the system clock in the default time-zone.

This will query the system clock in the default time-zone to obtain the current date.

Using this method will prevent the ability to use an alternate clock for testing because the clock is hard-coded.

Specified by:

dateNow in interface Chronology

Returns:

the current local date using the system clock and default time-zone, not null

dateNow

public MinguoDate dateNow(ZoneId zone)

Description copied from interface: Chronology

Obtains the current local date in this chronology from the system clock in the specified time-zone.

This will query the system clock to obtain the current date. Specifying the time-zone avoids dependence on the default time-zone.

Using this method will prevent the ability to use an alternate clock for testing because the clock is hard-coded.

Specified by:

dateNow in interface Chronology

Parameters:

zone - the zone ID to use, not null

Returns:

the current local date using the system clock, not null

dateNow

```
public MinguoDate dateNow(Clock clock)
```

Description copied from interface: Chronology
Obtains the current local date in this chronology from the specified clock.

This will query the specified clock to obtain the current date - today. Using this method allows the use of an alternate clock for testing. The alternate clock may be introduced using dependency injection.

Specified by:

dateNow in interface Chronology

Parameters:

clock - the clock to use, not null

Returns:

the current local date, not null

date

```
public MinguoDate date(TemporalAccessor temporal)
```

Description copied from interface: Chronology
Obtains a local date in this chronology from another temporal object.

This obtains a date in this chronology based on the specified temporal. A TemporalAccessor represents an arbitrary set of date and time information, which this factory converts to an instance of ChronoLocalDate.

The conversion typically uses the EPOCH_DAY field, which is standardized across calendar systems.

This method matches the signature of the functional interface TemporalQuery allowing it to be used as a query via method reference, aChronology::date.

Specified by:

date in interface Chronology

Parameters:

temporal - the temporal object to convert, not null

Returns:

the local date in this chronology, not null

See Also:

ChronoLocalDate.from(TemporalAccessor)

localDateTime

public ChronoLocalDateTime<MinguoDate> localDateTime(TemporalAccessor temporal)

Description copied from interface: Chronology

Obtains a local date-time in this chronology from another temporal object.

This obtains a date-time in this chronology based on the specified temporal. A TemporalAccessor represents an arbitrary set of date and time information, which this factory converts to an instance of ChronoLocalDateTime.

The conversion extracts and combines the ChronoLocalDate and the LocalTime from the temporal object. Implementations are permitted to perform optimizations such as accessing those fields that are equivalent to the relevant objects. The result uses this chronology.

This method matches the signature of the functional interface TemporalQuery allowing it to be used as a query via method reference, aChronology::localDateTime.

Specified by:

localDateTime in interface Chronology

Parameters:

temporal - the temporal object to convert, not null

Returns:

the local date-time in this chronology, not null

See Also:

ChronoLocalDateTime.from(TemporalAccessor)

zonedDateTime

```
public ChronoZonedDateTime<MinguoDate> zonedDateTime(TemporalAccessor temporal)
```

Description copied from interface: Chronology

Obtains a ChronoZonedDateTime in this chronology from another temporal object.

This obtains a zoned date-time in this chronology based on the specified temporal. A TemporalAccessor represents an arbitrary set of date and time information, which this factory converts to an instance of ChronoZonedDateTime.

The conversion will first obtain a ZoneId from the temporal object, falling back to a ZoneOffset if necessary. It will then try to obtain an Instant, falling back to a ChronoLocalDateTime if necessary. The result will be either the combination of ZoneId or ZoneOffset with Instant or ChronoLocalDateTime. Implementations are permitted to perform optimizations such as accessing those fields that are equivalent to the relevant objects. The result uses this chronology.

This method matches the signature of the functional interface TemporalQuery allowing it to be used as a query via method reference, aChronology::zonedDateTime.

Specified by:

zonedDateTime in interface Chronology

Parameters:

temporal - the temporal object to convert, not null

Returns:

the zoned date-time in this chronology, not null

See Also:

ChronoZonedDateTime.from(TemporalAccessor)

zonedDateTime

```
public ChronoZonedDateTime<MinguoDate> zonedDateTime(Instant instant,
                                                     ZoneId zone)
```

Description copied from interface: Chronology

Obtains a ChronoZonedDateTime in this chronology from an Instant.

This obtains a zoned date-time with the same instant as that specified.

Specified by:

zonedDateTime in interface Chronology

Parameters:

`instant - the instant to create the date-time from, not null`

`zone - the time-zone, not null`

Returns:

`the zoned date-time, not null`

isLeapYear

```
public boolean isLeapYear(long prolepticYear)
```

Checks if the specified year is a leap year.

Minguo leap years occur exactly in line with ISO leap years. This method does not validate the year passed in, and only has a well-defined result for years in the supported range.

Specified by:

`isLeapYear` in interface `Chronology`

Parameters:

`prolepticYear - the proleptic-year to check, not validated for range`

Returns:

`true if the year is a leap year`

prolepticYear

```
public int prolepticYear(Era era,
                         int yearOfEra)
```

Description copied from interface: `Chronology`

Calculates the proleptic-year given the era and year-of-era.

This combines the era and year-of-era into the single proleptic-year field.

If the chronology makes active use of eras, such as `JapaneseChronology` then the year-of-era will be validated against the era. For other chronologies, validation is optional.

Specified by:

`prolepticYear` in interface `Chronology`

Parameters:

`era - the era of the correct type for the chronology, not null`

`yearOfEra - the chronology year-of-era`

Returns:

the proleptic-year

eraOf

public MinguoEra eraOf(int eraValue)

Description copied from interface: Chronology
Creates the chronology era object from the numeric value.

The era is, conceptually, the largest division of the time-line. Most calendar systems have a single epoch dividing the time-line into two eras. However, some have multiple eras, such as one for the reign of each leader. The exact meaning is determined by the chronology according to the following constraints.

The era in use at 1970-01-01 must have the value 1. Later eras must have sequentially higher values. Earlier eras must have sequentially lower values. Each chronology must refer to an enum or similar singleton to provide the era values.

This method returns the singleton era of the correct type for the specified era value.

Specified by:

eraOf in interface Chronology

Parameters:

eraValue - the era value

Returns:

the calendar system era, not null

eras

public List<Era> eras()

Description copied from interface: Chronology
Gets the list of eras for the chronology.

Most calendar systems have an era, within which the year has meaning. If the calendar system does not support the concept of eras, an empty list must be returned.

Specified by:

eras in interface Chronology

Returns:

the list of eras for the chronology, may be immutable, not null

range

```
public ValueRange range(ChronoField field)
```

Description copied from interface: Chronology

Gets the range of valid values for the specified field.

All fields can be expressed as a `long` integer. This method returns an object that describes the valid range for that value.

Note that the result only describes the minimum and maximum valid values and it is important not to read too much into them. For example, there could be values within the range that are invalid for the field.

This method will return a result whether or not the chronology supports the field.

Specified by:

`range` in interface `Chronology`

Parameters:

`field - the field to get the range for, not null`

Returns:

`the range of valid values for the field, not null`

resolveDate

```
public MinguoDate resolveDate(Map<TemporalField,Long> fieldValues,
                              ResolverStyle resolverStyle)
```

Description copied from class: AbstractChronology

Resolves parsed `ChronoField` values into a date during parsing.

Most `TemporalField` implementations are resolved using the resolve method on the field. By contrast, the `ChronoField` class defines fields that only have meaning relative to the chronology. As such, `ChronoField` date fields are resolved here in the context of a specific chronology.

`ChronoField` instances are resolved by this method, which may be overridden in subclasses.

- `EPOCH_DAY` - If present, this is converted to a date and all other date fields are then cross-checked against the date.
- `PROLEPTIC_MONTH` - If present, then it is split into the `YEAR` and `MONTH_OF_YEAR`.
  If the mode is strict or smart then the field is validated.
- `YEAR_OF_ERA` and `ERA` - If both are present, then they are combined to form a `YEAR`. In lenient mode, the `YEAR_OF_ERA` range is not validated, in smart and strict mode it is. The `ERA` is validated for range in all three

modes. If only the YEAR_OF_ERA is present, and the mode is smart or lenient, then the last available era is assumed. In strict mode, no era is assumed and the YEAR_OF_ERA is left untouched. If only the ERA is present, then it is left untouched.

- YEAR, MONTH_OF_YEAR and DAY_OF_MONTH - If all three are present, then they are combined to form a date. In all three modes, the YEAR is validated. If the mode is smart or strict, then the month and day are validated. If the mode is lenient, then the date is combined in a manner equivalent to creating a date on the first day of the first month in the requested year, then adding the difference in months, then the difference in days. If the mode is smart, and the day-of-month is greater than the maximum for the year-month, then the day-of-month is adjusted to the last day-of-month. If the mode is strict, then the three fields must form a valid date.
- YEAR and DAY_OF_YEAR - If both are present, then they are combined to form a date. In all three modes, the YEAR is validated. If the mode is lenient, then the date is combined in a manner equivalent to creating a date on the first day of the requested year, then adding the difference in days. If the mode is smart or strict, then the two fields must form a valid date.
- YEAR, MONTH_OF_YEAR, ALIGNED_WEEK_OF_MONTH and ALIGNED_DAY_OF_WEEK_IN_MONTH - If all four are present, then they are combined to form a date. In all three modes, the YEAR is validated. If the mode is lenient, then the date is combined in a manner equivalent to creating a date on the first day of the first month in the requested year, then adding the difference in months, then the difference in weeks, then in days. If the mode is smart or strict, then the all four fields are validated to their outer ranges. The date is then combined in a manner equivalent to creating a date on the first day of the requested year and month, then adding the amount in weeks and days to reach their values. If the mode is strict, the date is additionally validated to check that the day and week adjustment did not change the month.
- YEAR, MONTH_OF_YEAR, ALIGNED_WEEK_OF_MONTH and DAY_OF_WEEK - If all four are present, then they are combined to form a date. The approach is the same as described above for years, months and weeks in ALIGNED_DAY_OF_WEEK_IN_MONTH. The day-of-week is adjusted as the next or same matching day-of-week once the years, months and weeks have been handled.
- YEAR, ALIGNED_WEEK_OF_YEAR and ALIGNED_DAY_OF_WEEK_IN_YEAR - If all three are present, then they are combined to form a date. In all three modes, the YEAR is validated. If the mode is lenient, then the date is combined in a manner equivalent to creating a date on the first day of the requested year, then adding the difference in weeks, then in days. If the mode is smart or strict, then the all three fields are validated to their outer ranges. The date is then combined in a manner equivalent to creating a date on the first day of the requested year, then adding the amount in weeks and days to reach their values. If the mode is strict, the date is additionally validated to check that the day and week adjustment did not change the year.

- `YEAR`, `ALIGNED_WEEK_OF_YEAR` and `DAY_OF_WEEK` - If all three are present, then they are combined to form a date. The approach is the same as described above for years and weeks in `ALIGNED_DAY_OF_WEEK_IN_YEAR`. The day-of-week is adjusted as the next or same matching day-of-week once the years and weeks have been handled.

The default implementation is suitable for most calendar systems. If `ChronoField.YEAR_OF_ERA` is found without an `ChronoField.ERA` then the last era in `Chronology.eras()` is used. The implementation assumes a 7 day week, that the first day-of-month has the value 1, that first day-of-year has the value 1, and that the first of the month and year always exists.

Specified by:

`resolveDate` in interface `Chronology`

Overrides:

`resolveDate` in class `AbstractChronology`

Parameters:

`fieldValues` - the map of fields to values, which can be updated, not null

`resolverStyle` - the requested type of resolve, not null

Returns:

the resolved date, null if insufficient information to create a date

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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